

Infosys Springboard Virtual Internship 6.0 Completion Report

Batch Number: 13

Start date: February 5, 2026

Name: Rajesh Yadav

Internship Duration: 8 Weeks

Project Mentor: N.Shanmukha Priya,

1. Project Title: ClimateScope: Visualizing Global Weather Trends and Extreme Events

2. Project Objective:

To analyze global weather data and create interactive visualizations that highlight trends, regional variations, and extreme weather events, enabling better understanding and data-driven climate insights

3. Project description in detail:

The ClimateScope project is designed to analyze and visualize global weather data using an interactive dashboard. It leverages the Global Weather Repository dataset to explore key weather parameters such as temperature, humidity, rainfall, wind speed, and air quality. The dashboard provides insights into monthly temperature and humidity trends, helping identify seasonal patterns and variations.

It also includes a weather condition distribution, showcasing the frequency of conditions like sunny, cloudy, and rainy, enabling a better understanding of global climate behavior. Key metrics such as average temperature, average humidity, and total rainfall offer a quick overview of overall weather conditions. Additionally, location-based analysis highlights differences in temperature and air quality (PM2.5 and PM10) across cities, helping identify regions with extreme conditions. Advanced visualizations like scatter plots (wind speed vs temperature) and UV index analysis reveal relationships between weather factors, while precipitation and monthly trend charts provide deeper insights into climate changes over time.

The dashboard is interactive, allowing users to filter data by location, country, and weather condition, making it easy to explore and compare different regions.

Overall, the project transforms raw weather data into meaningful insights, supporting climate awareness, analysis, and data-driven decision-making.

4. Timeline Overview:

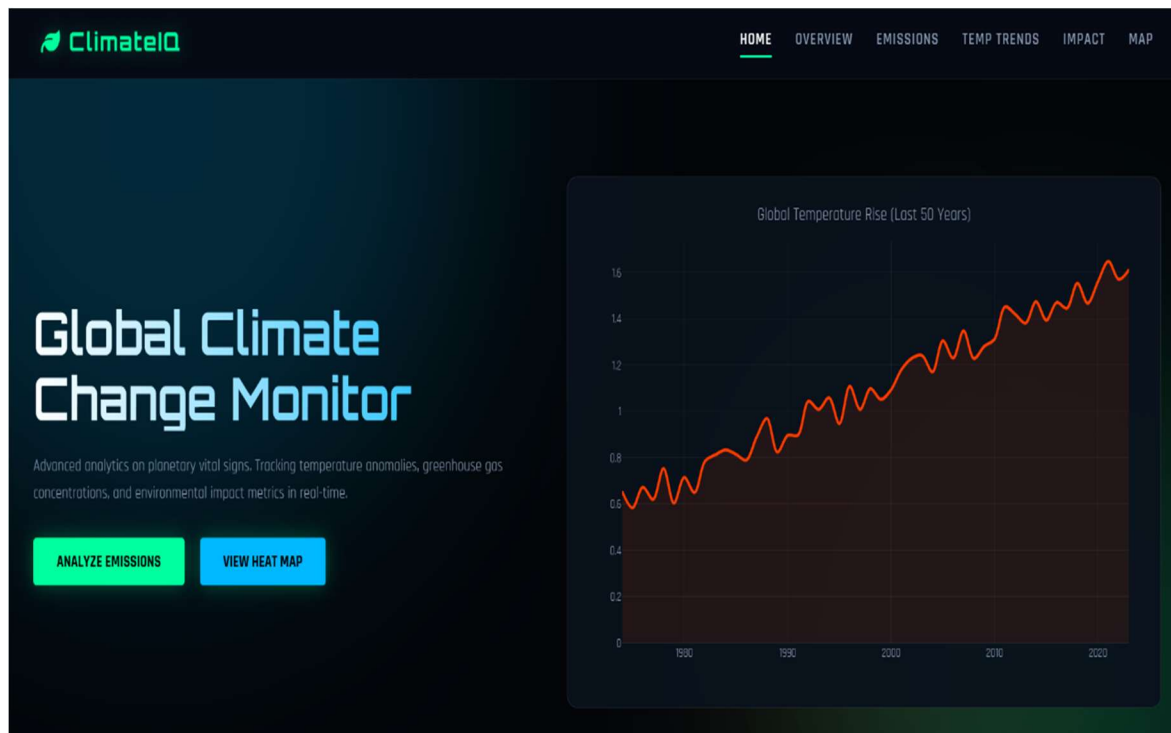
Week	Activities Planned	Activities Completed
Week 1	Data Collection & Understanding	Download Global Weather Repository dataset (Kaggle) Explore dataset structure and variables (temperature, humidity, etc.) Identify missing values and anomalies.
Week 2	Data Cleaning & Preprocessing	Handle missing and inconsistent data Convert units and normalize values Prepare dataset for analysis (filtering, aggregation like monthly averages).
Week 3	Data Analysis	Perform statistical analysis (mean, trends, correlations) Identify seasonal patterns and extreme weather events Compare weather conditions across different regions
Week 4	Visualization Design	Decide chart types (line chart, bar chart, heatmap, maps), Design dashboard layout (wireframe/mockup) ,Plan KPIs (avg temperature, humidity, rainfall)
Week 5	Dashboard Development	Create visualizations in Power BI / Tableau / Plotly Build charts: temperature trends, air quality, weather distribution, etc. Add interactivity (filters, slicers, dropdowns)
Week 6	Advanced Analysis & Insights	Add advanced visuals (scatter plots, UV index, rainfall trends) Generate key insights and observations Highlight anomalies and patterns
Week 7	Testing & Review	Test dashboard functionality and accuracy Improve UI/UX (labels, colors, layout) Fix errors and optimize performance
Week 8	Documentation	Prepare final project report Summarize methodology and insights Deploy dashboard, prepare presentation.

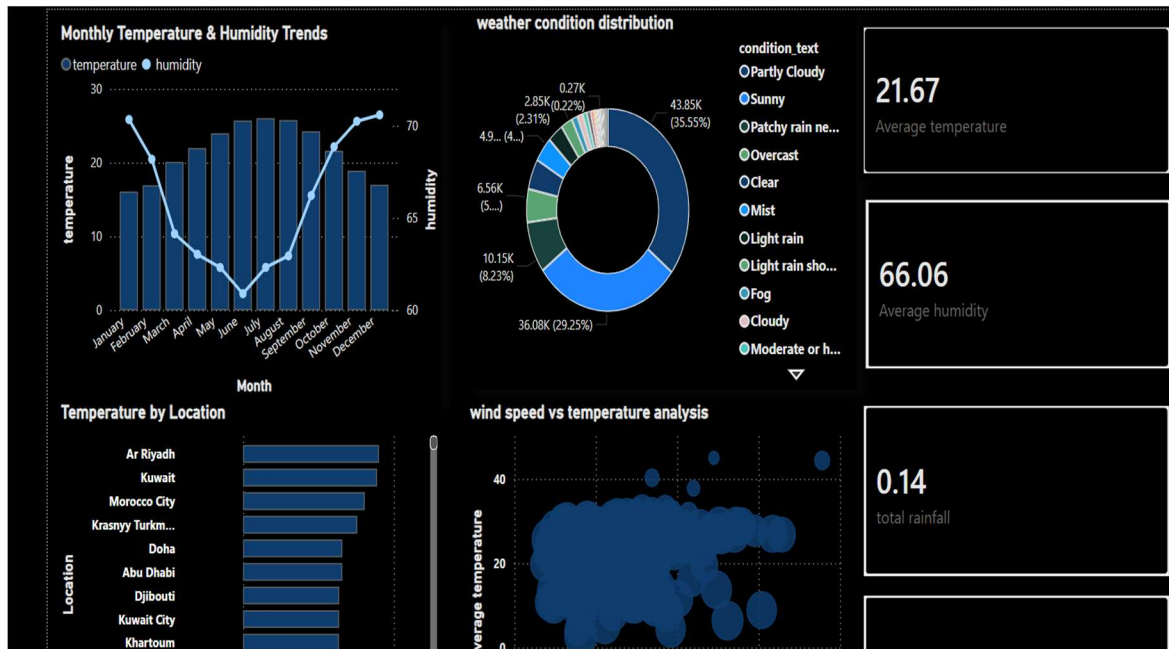
5a. Key Milestones

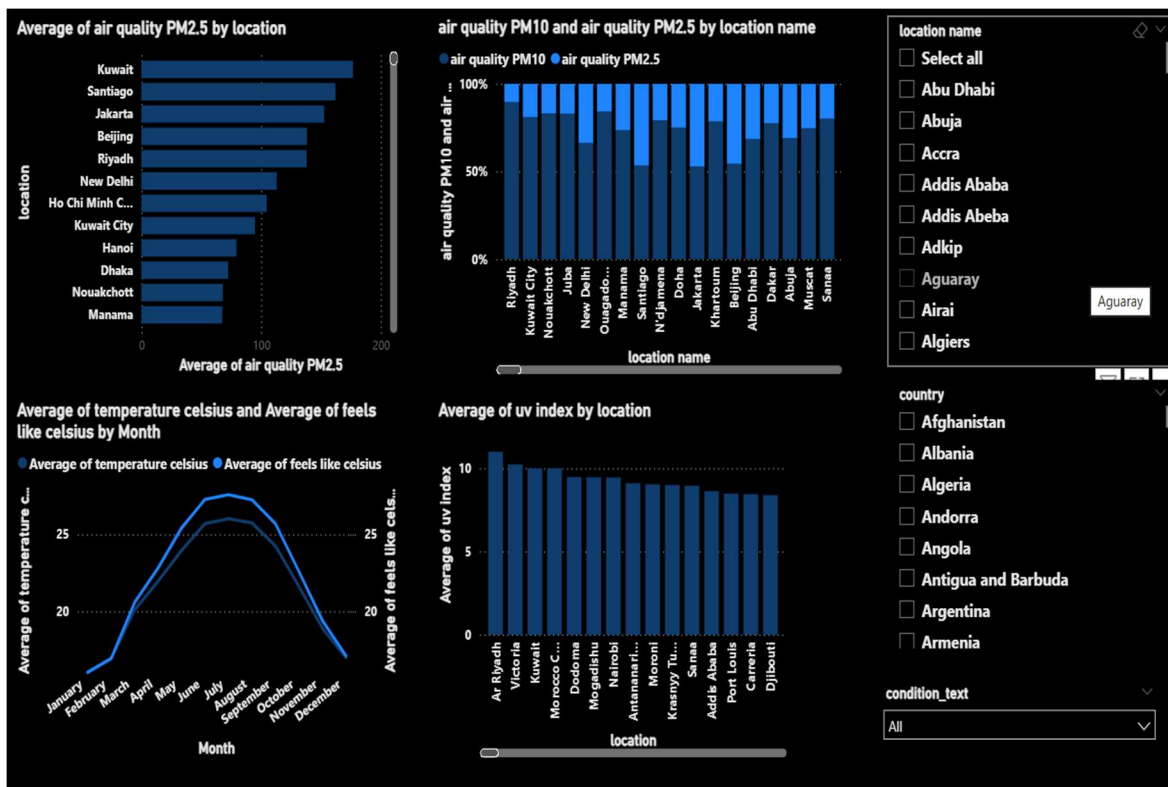
Milestone	Description	Date Achieved
Project Kickoff	Internship start date	February 5, 2026
Prototype/First Draft	Data cleaning, preprocessing, and initial dashboard creation in Power BI	February 19, 2026
Mid-Term Review	Implementation of visualizations (trends, air quality, weather distribution)	March 13, 2026
Final Submission	Final dashboard completion with insights and testing	March 27, 2026
Presentation	Completion of documentation and final delivery	End of March 2026

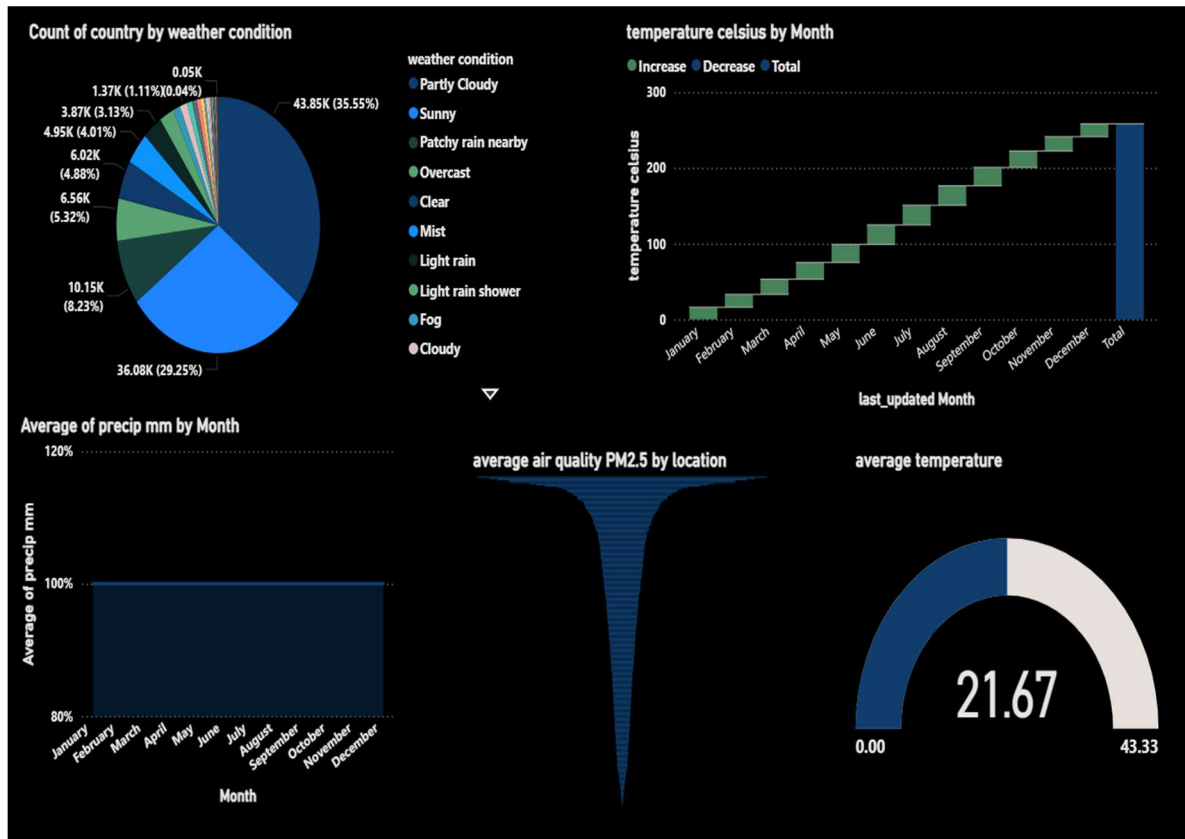
5b. Project execution details:

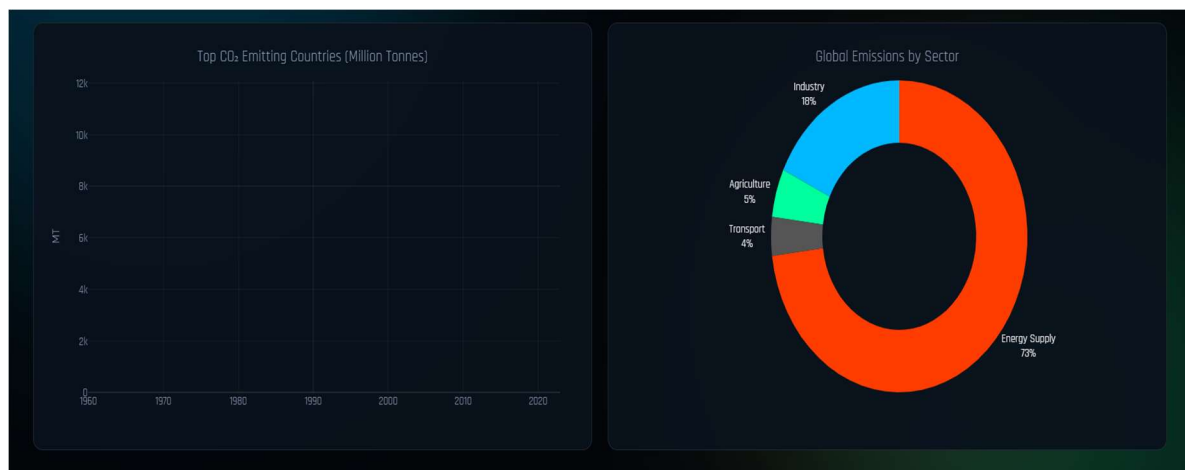
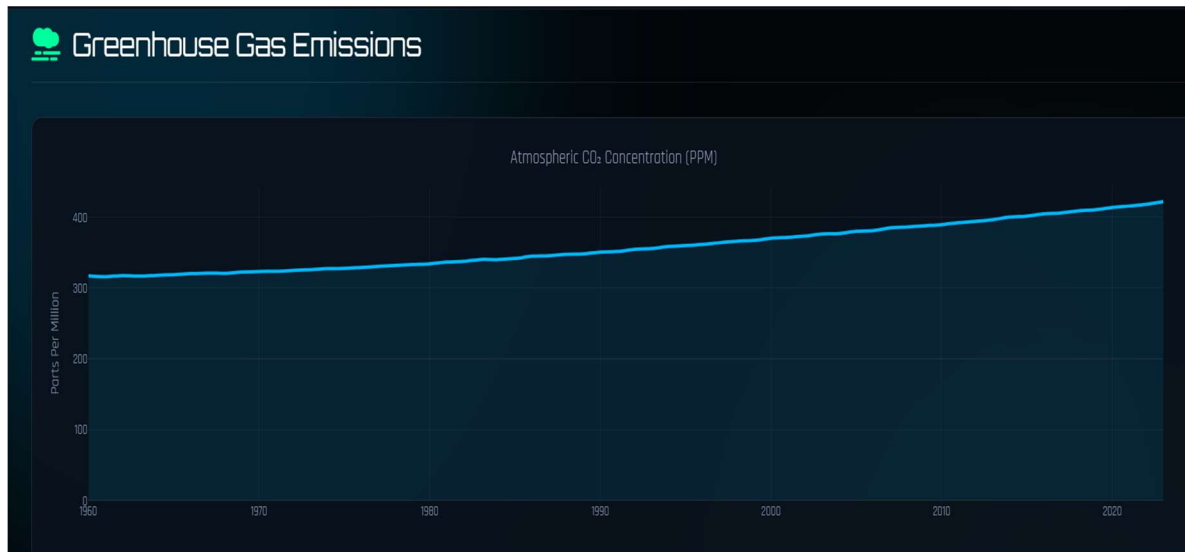
The project began with collecting and understanding the Global Weather dataset. Data cleaning and preprocessing were performed to handle missing values and prepare the data for analysis. Exploratory analysis was conducted to identify trends, seasonal patterns, and correlations. An interactive dashboard was then developed using Power BI, incorporating visualizations such as charts, KPIs, and scatter plots. Filters and slicers were added for better user interaction. Finally, the dashboard was tested, refined, and used to generate key insights for reporting and presentation.

6. Snapshots / Screenshots









7. Challenges Faced

- Handling missing and inconsistent data in the dataset
- Managing large datasets, which affected performance during processing
- Choosing the right visualizations to represent complex weather data effectively
- Dealing with data variability across different locations and time periods
- Implementing interactive features (filters, slicers) smoothly in the dashboard
- Ensuring accuracy and clarity of insights while combining multiple parameters (temperature, humidity, air quality, etc.).

8. Learnings & Skills Acquired: Through this project, I gained practical experience in data preprocessing and cleaning, including handling missing and inconsistent data. I developed strong skills in exploratory data analysis (EDA) to identify trends, patterns, and correlations in real-world datasets. I enhanced my ability to create interactive dashboards using Power BI, including the use of charts, KPIs, filters, and slicers. Additionally, I improved my understanding of data visualization techniques and selecting appropriate charts for different types of data. The project also strengthened my skills in analytical thinking, problem-solving, and data interpretation, along with gaining insights into climate data analysis and environmental patterns.

9. Testimonials from team: The team members appreciated the collaborative efforts and effective communication throughout the project. Each member actively contributed to data analysis, visualization, and dashboard development, ensuring smooth execution of tasks. The project environment encouraged teamwork, problem-solving, and knowledge sharing, which helped in overcoming challenges efficiently. The coordination among team members played a key role in delivering a well-structured and insightful dashboard. Overall, the team expressed satisfaction with the project outcome, highlighting the practical learning experience, skill development, and successful implementation of data analytics and visualization techniques.

10. Conclusion: The The ClimateScope project successfully transformed raw global weather data into meaningful and interactive visual insights. Through data preprocessing, analysis, and dashboard development, the project effectively highlighted climate trends, regional variations, and key weather patterns. The interactive dashboard enables users to explore data dynamically, making it easier to understand complex weather information and derive actionable insights. Overall, the project achieved its objective of enhancing climate awareness and supporting data-driven decision-making, while also providing valuable hands-on experience in data analytics and visualization.

11. Acknowledgements: I would like to express my sincere gratitude to my mentor for their valuable guidance, support, and encouragement throughout the project. Their insights and suggestions played a crucial role in the successful completion of this work. I also thank my team members for their cooperation, teamwork, and continuous support during all phases of the project. Additionally, I am grateful to the organization and faculty for providing the necessary resources and opportunity to work on this project. Finally, I would like to acknowledge all the online resources and tools that contributed to the successful execution of this project.